SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Rural Branch Office – Internet Connectivity and Security

1. Understanding of Problem & Constraints

The company is opening a branch office in a remote rural area where high-speed wired internet is unavailable.

Main constraints:

- Wired broadband is unavailable.
- Cloud applications require stable internet.
- Video conferencing requires sufficient bandwidth and low latency.
- Secure communication with headquarters is essential.
- Budget is limited.
- Business operations should continue even if the primary connection fails.
- The network should support future expansion.

2. Problem Decomposition

The problem can be divided into four parts:

1. Select a suitable internet connection.
2. Select networking devices.
3. Provide security for communication.
4. Provide backup connectivity and future scalability.

3. Proposed Solution

Internet connection:

- Use **4G/5G fixed wireless broadband** as the primary connection where available.
- If cellular coverage is poor, use **satellite internet** as an alternative.
- Use a **dual-WAN router** to support automatic failover between two connections.

Suggested network:

Internet → Dual-WAN Router → Firewall → Network Switch → PCs / Wi-Fi Access Points

4. Networking Devices

Device	Purpose
4G/5G modem/router	Primary internet connection
Satellite connection	Backup connectivity

Device	Purpose
Dual-WAN router	Automatic connection switching
Firewall	Network security
Managed switch	Connects wired devices
Wi-Fi 6 access point	Wireless connectivity
UPS	Keeps networking equipment running during short power failures

5. Security Mechanisms

- Firewall to block unauthorized traffic.
- **VPN** between branch and headquarters.
- WPA3 for Wi-Fi security.
- Strong user authentication.
- Network segmentation for employees and guests.
- Regular firmware and security updates.
- IDS/IPS where budget permits.

6. Constraint-Based Reasoning

Bandwidth:
4G/5G can provide adequate bandwidth for cloud applications and video meetings. Video traffic can be prioritized using **QoS**.

Reliability:
A second connection provides backup. If the primary connection fails, the dual-WAN router automatically switches to the backup.

Cost:
Using 4G/5G as the primary connection is generally more economical than deploying dedicated infrastructure in a remote area. Satellite can be reserved as backup if affordable.

Security:
A VPN provides encrypted communication with headquarters, while the firewall and WPA3 protect the local network.

Future scalability:
A managed switch and Wi-Fi 6 infrastructure allow additional users and devices to be added later.

7. Innovative Approach

Use **automatic WAN failover + QoS**:

- Critical business traffic gets priority.
- Non-critical traffic can be restricted during low bandwidth.

- Backup internet activates automatically.

8. Final Recommendation

A **4G/5G connection with a dual-WAN router, firewall, VPN, managed switch, Wi-Fi 6 access points, and backup satellite/secondary cellular connection** provides the best balance between **cost, reliability, security, bandwidth, and scalability**.

Expected outcome: Reliable cloud access, secure headquarters communication, uninterrupted operations, and an easily expandable network.

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2. University Campus – Wireless Network for 3,000+ Students

1. Understanding of Problem & Constraints

The university needs campus-wide Wi-Fi for more than **3,000 students** in classrooms, laboratories, libraries, and hostels.

Main constraints:

- More than 3,000 users.
- Many simultaneous connections.
- Limited number of access points.
- Limited budget.
- Signal interference must be minimized.
- Secure authentication is required.
- Good coverage and performance are required.

2. Problem Decomposition

The problem can be divided into:

1. Access point placement.
2. Frequency-band selection.
3. Capacity and interference management.
4. Authentication and security.
5. Cost and scalability.

3. Proposed Network Design

Use **Wi-Fi 6 (802.11ax)** access points.

Basic architecture:

Internet → Core Router/Firewall → Managed Switches → Wi-Fi 6 APs → Students/Staff

4. Access Point Placement

Access points should be strategically placed in:

- Classrooms.
- Laboratories.
- Library.
- Hostels.
- Common areas.
- High-density student locations.

Instead of placing APs randomly, perform a **site survey** to identify dead zones and interference.

High-density areas should receive more AP capacity, while low-density areas can use fewer APs.

5. Frequency Band Selection

2.4 GHz:

- Longer range.
- Better penetration through walls.
- More interference.
- Fewer non-overlapping channels.

5 GHz:

- Higher speed.
- More channels.
- Less interference.
- Shorter range.

6 GHz, if supported by the institution's Wi-Fi 6E equipment and local regulatory requirements, can provide additional spectrum for high-density environments.

Recommendation: Prefer **5 GHz/6 GHz for high-density users** and use 2.4 GHz mainly for devices requiring longer range or legacy compatibility.

6. Authentication and Security

Use:

- **WPA3-Enterprise** where supported.
- 802.1X authentication.
- RADIUS server for centralized authentication.
- Separate networks/VLANs for students, faculty, guests, and administrative systems.
- Strong encryption.
- Guest access isolated from internal resources.

7. Constraint-Based Reasoning

Coverage:
APs are placed according to a site survey rather than simply using equal spacing.

Performance:
Wi-Fi 6 improves performance in high-density environments using technologies such as **OFDMA and MU-MIMO**.

Interference:
Proper channel planning and reduced transmit power in overlapping areas help minimize interference.

Security:
WPA3-Enterprise and 802.1X prevent unauthorized users from accessing university resources.

Cost:
Since the number of APs is limited, APs should be concentrated in high-density areas instead of deploying the same number everywhere.

8. Innovative Approach

Use **capacity-based AP deployment** rather than only coverage-based deployment.

For example:

- Library → High AP density.
- Classroom blocks → Medium/high density.
- Hostels → High density during peak hours.
- Low-use outdoor areas → Lower AP density.

This makes better use of the limited AP budget.

9. Final Recommendation

A **Wi-Fi 6 campus network with strategically positioned APs, 5 GHz/6 GHz prioritized for high-density users, 2.4 GHz for longer-range/legacy devices, WPA3-Enterprise, 802.1X, RADIUS, and VLAN segmentation** provides the best balance between **coverage, performance, security, and cost**.

Expected outcome: Reliable campus-wide connectivity for 3,000+ users with reduced interference and secure access.

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3. Financial Institution – Secure Internal Network

1. Understanding of Problem & Constraints

The financial institution handles sensitive customer and banking information, so cybersecurity is extremely important.

Main constraints:

- Customer data must be protected.
- Employees need access to sensitive systems.
- Online banking services must remain available.
- Strict security policies must be followed.
- Network performance should not decrease significantly.
- Operational costs should remain controlled.
- Security monitoring must be continuous.

2. Problem Decomposition

The solution can be divided into:

1. Network segmentation.
2. Firewall deployment.
3. Authentication.
4. Intrusion detection and prevention.
5. Monitoring and compliance.
6. Performance and cost optimization.

3. Proposed Network Architecture

A layered architecture is recommended:

Internet

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Edge Firewall

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DMZ

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Internal Firewall

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Core Network

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VLANs

- Employee VLAN
- Customer-data VLAN
- Banking application VLAN
- Server VLAN
- Management VLAN
- Guest VLAN

4. Network Segmentation

Use **VLANs and firewall-controlled zones**.

For example:

Network	Purpose
Employee VLAN	Normal employee systems
Server VLAN	Internal servers
Customer Data VLAN	Sensitive customer information
Banking VLAN	Banking applications
Management VLAN	Network administration
Guest VLAN	Guest devices

Sensitive systems should not communicate directly with ordinary employee or guest networks.

5. Firewall Deployment

Use **next-generation firewalls (NGFW)** at the network perimeter.

Firewall functions include:

- Traffic filtering.
- Application control.
- Intrusion prevention.
- VPN security.
- Malware protection.
- Logging and monitoring.

A separate internal firewall or security zone can protect highly sensitive systems.

6. Authentication

Use:

- **Multi-factor authentication (MFA)**.
- 802.1X for network access.
- Role-based access control (RBAC).
- Strong passwords.
- Centralized identity management.

Employees should receive only the permissions required for their jobs.

7. Intrusion Detection

Deploy **IDS/IPS** to monitor suspicious activities.

Examples of activities detected:

- Unauthorized login attempts.

- Port scanning.
- Malware traffic.
- Abnormal data transfers.
- Suspicious internal communication.

Logs can be collected in a **SIEM system** for centralized monitoring and investigation.

8. Constraint-Based Reasoning

Security:
Segmentation, firewalls, MFA, IDS/IPS, and encryption provide multiple layers of protection.

Performance:
Traffic can be filtered close to its source, and network policies can prevent unnecessary traffic from reaching critical systems.

Compliance:
Access control, centralized logging, monitoring, and auditing support strict financial security requirements.

Budget:
Use existing network infrastructure where possible and prioritize security controls around critical systems rather than deploying expensive equipment everywhere.

9. Innovative Approach

Use a **Zero-Trust approach**:

"Never automatically trust a user or device; verify access before allowing it."

Even an employee inside the organization should be authenticated and authorized before accessing sensitive customer information.

10. Final Recommendation

A **segmented network using VLANs, next-generation firewalls, DMZ, MFA, 802.1X, RBAC, IDS/IPS, encryption, and centralized SIEM monitoring** provides strong security without unnecessarily affecting network performance.

Expected outcome: Sensitive customer information is protected, employees receive controlled access, suspicious activities are detected quickly, and the organization can meet strict security and compliance requirements.
